



NEXUS-1 — Project Brief

Nuclear Plant Operator Console & Digital-Twin Platform · Phase 0

One-look overview · Build NX1-094C79E0A366D34F · June 2026

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github.com/gregory82gr/Nexus-1-phase-0
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An interactive operator console and digital-twin demonstration for a nuclear power facility — a single self-contained web application (~41 screens across ~25 menu areas) driven by a live in-browser simulation. It presents an entire plant — physics, generation, safety, fleet, lifecycle, people and analytics — as one connected system: a SCRAM ripples into incidents, compliance, inspection demand and component ageing, rather than living on an isolated dashboard.

AT A GLANCE

Reactor & physics

Core map, control rods + SCRAM, neutronics; a real six-group point-kinetics solver with Doppler, moderator and xenon feedback; a Model Analysis screen verifying it against closed-form references; 3D reactor twin.

Fleet

Multi-unit operation — PWR-900 / PWR-1300 units and NUWARD-style SMR modules — with grid aggregation and a unit selector; every screen follows the selected unit.

Secondary & electrical

Steam generators, primary/secondary coolant loops, turbine-generator train, power and 400 kV grid synchronisation (50 Hz conventions).

Safety & monitoring

Radiation and dose, prioritised alarms, incident timeline, hash-chained tamper-evident audit log, platform & OT security view.

Intelligence

AI predictive diagnostics; RL optimiser (twin-only, advisory, readable policy); dependency graph / influence matrix / causal chain; a DSLM assistant (local Ollama, air-gapped) documented as ROADMAP · PLANNED.

Root-cause analysis

An engineered fault-propagation graph lit live by alarms: an alarm flood of ~14 cascading alarms collapses to one causal origin, with causal ranking pointing upstream where naive alarm counts point at symptoms.

Training

Training Mode on the live kinetics model — five scored drills (power maneuver, deep reduction, trip response, xenon transient, load-follow) with objectives, live scoring and a logged debrief.

Lifecycle, people & access

NDT rod inspection, component registry with live health, ageing & degradation, decommissioning, waste & spent fuel, staffing + stress test, robotic fleet + mission readiness, zone access.

WHAT IS REAL VS. WHAT IS SIMULATED

REAL / GENUINE

The interface, navigation, visualisations and interactions are working code.

The point-kinetics solver and the RL agent are real in structure and genuinely compute / learn.

The Neutronics tab uses honest, clearly-labelled display mappings driven by the fleet model.

Workflows reflect real industry concepts — NDT methods, dose limits, decommissioning stages.

SIMULATED / ILLUSTRATIVE

All data is generated in the browser; nothing connects to a real plant.

Their constants are representative textbook-order values, not tuned to a specific reactor.

Its detector readings and reactivity-balance rows are illustrative; they do not run the kinetics solver.

Ageing figures, dependency weights, names, ratings and dates are plausible placeholders.

OPERATING MODES — DESTINATION: ALL SIX

● Simulation

● Training

Shadow

Advisory

Supervisory

Hybrid Digital Twin

Phase 0 ships Simulation and Training as working modes; Shadow, Advisory, Supervisory and Hybrid Digital Twin are roadmap.

Invariant in every mode: no control authority. The platform watches, trains, advises and reconciles — it never actuates plant equipment. Certified reactor-protection functions remain hardware-isolated, deterministic and outside this system; any real-plant link is read-only, proven by network policy and code audit, not asserted.

DESTINATION & TECHNOLOGY (PER ROADMAP V2 DECISION RECORDS)

.NET C# microservices (no Orleans) · REST / OpenAPI behind a YARP gateway · Angular · SignalR (Redis backplane) · Kafka as the single messaging substrate — events, commands and telemetry · PostgreSQL per service + TimescaleDB historian · Keycloak (OIDC, RBAC) · Docker / Kubernetes, air-gap deployable · OpenTelemetry → Prometheus / Grafana / Loki

Build-out is sequenced vertical-slice-first: milestone M1 (~750-950 h) proves the Kafka + SignalR + historian spine end-to-end before broad spend. Planning estimates: ~6,950 senior-engineer hours for the core platform (Target A), ~8,350 h with all six modes engineered, ~10-18 months for a small-to-balanced team. Regulated, plant-connected deployment is Track B — a separate, specialist-led nuclear V&V programme (12-36+ months), deliberately not estimated as software engineering.

By Grigorios Agathangelidis (Electrical & Software Engineering) — a personal project applying software architecture, simulation, visualisation and systems design to a studied domain; not a claim of nuclear-engineering expertise.

Demonstration and educational tool: not a licensed analysis tool and not connected to any real reactor. Every displayed number traces to something the code computes or to a labelled illustrative mapping.

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